

Career Data-Scraper

Introduction

Data Analytics involves collecting, processing, analyzing, and visualizing data to generate meaningful insights. This project demonstrates a complete analytics workflow by collecting job listing data from a public job portal, performing exploratory data analysis, and creating visualizations to understand hiring trends.

Project Objective

The primary objective of this project was to:

- Collect job listing data using web scraping.
- Store the data in a structured CSV format.
- Perform exploratory data analysis (EDA).
- Identify trends and patterns in job postings.
- Create visualizations for better understanding of the data.

Task 1: Web Scraping

Description

A Python-based web scraper was developed to extract job listing information from a public job portal.

Tools Used

- Python
- Requests
- BeautifulSoup
- Pandas

Data Collected

The scraper collected the following fields:

- Job Title
- Company Name
- Location

Output

The extracted data was stored in a CSV file named:

jobs.csv

This dataset served as the foundation for further analysis.

Task 2: Exploratory Data Analysis (EDA)

Objective

To understand the structure and quality of the collected dataset and identify meaningful trends.

Analysis Performed

Dataset Exploration

The dataset was loaded using Pandas and inspected to understand:

- Number of rows
- Number of columns
- Data types
- Dataset structure

Missing Value Analysis

The dataset was checked for missing values to ensure data quality.

Duplicate Record Analysis

Duplicate records were identified and evaluated.

Frequency Analysis

The following analyses were performed:

- Most common job titles
- Most active hiring companies
- Most common job locations

Hypothesis

Hypothesis:

Job opportunities are concentrated in a limited number of locations and job roles.

Result

The analysis supported this hypothesis, as a few locations and job titles appeared significantly more frequently than others.

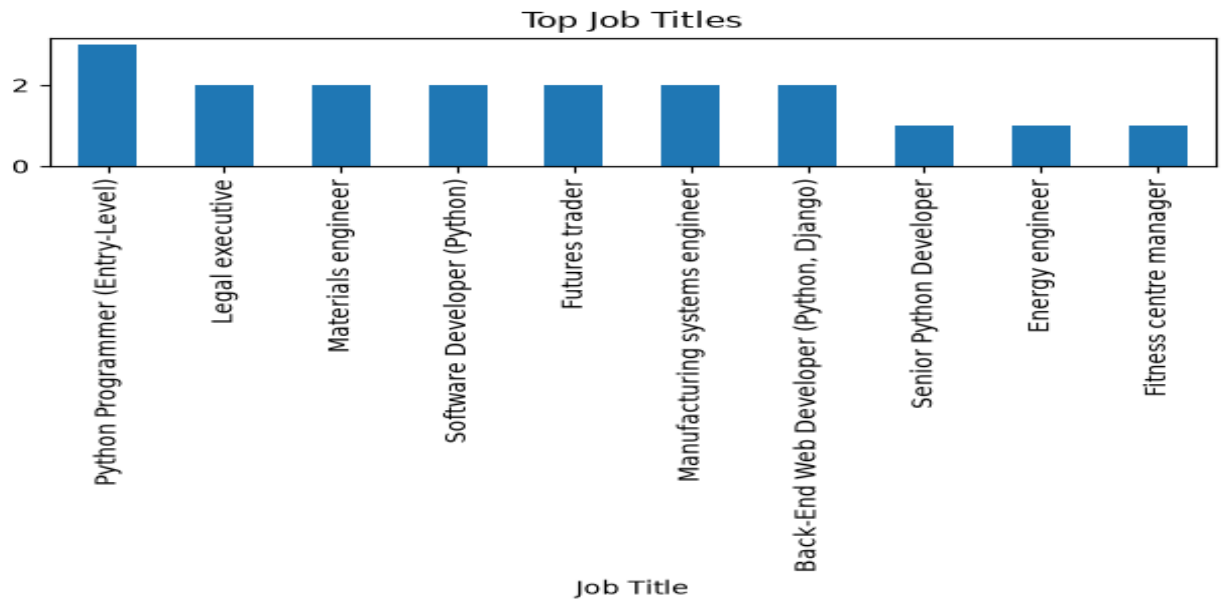
Task 3: Data Visualization

Objective

To transform raw data into visual insights that are easier to understand and communicate.

Visualizations Created

1. Top Job Titles



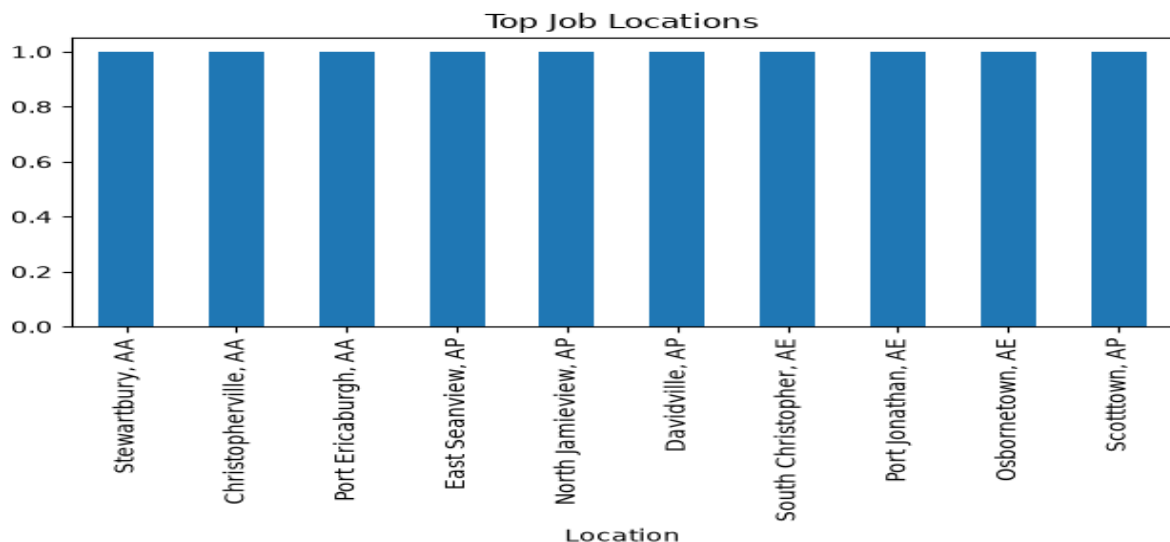
A bar chart was created to identify the most frequently occurring job roles in the dataset.

2. Top Hiring Companies



A bar chart was created to identify companies with the highest number of job postings.

3. Top Job Locations



A bar chart was created to identify locations with the highest concentration of job opportunities.

Data Storytelling

The visualizations revealed several important insights:

- Job opportunities were concentrated in a few major locations.
- Certain job roles appeared significantly more frequently than others.
- A limited number of companies accounted for a large portion of job postings.
- Hiring activity was not evenly distributed across locations.

These findings indicate that employment opportunities tend to cluster around major business and technology hubs, highlighting important trends in the job market.

Key Findings

1. The dataset successfully captured job market information from a public job portal.
2. Certain job titles dominated the dataset.
3. A small group of companies contributed a significant number of job postings.
4. Major locations showed higher hiring activity compared to other regions.
5. Visual analysis made hiring trends easier to understand.

Conclusion

This project successfully demonstrated the complete data analytics workflow, beginning with web scraping, followed by exploratory data analysis, and ending with data visualization.

The project provided valuable insights into job market trends while strengthening practical skills in Python, data collection, data analysis, and visualization.

Technologies Used

- Python
- Requests
- BeautifulSoup
- Pandas
- Matplotlib
- Git
- GitHub

Project Files

- job_scraper.py
- eda_analysis.py
- visualization.py
- jobs.csv
- README.md
- Visualization Images

Future Improvements

- Collect larger datasets from multiple job portals.
- Include salary and experience information.
- Build an interactive dashboard using Power BI or Tableau.
- Automate periodic data collection and reporting.

End of Report